

ANDREJ ORSULA

Researcher in AI for Space Robotics

Luxembourg

andrej.orsula@uni.lu ♦ andrejorsula.github.io ♦ github.com/andrejorsula

EXPERIENCE

Frontier Development Lab

Researcher

Italy & Online

Jun 2026 – Present

- Development of a multimodal foundation models for the lunar south pole

University of Luxembourg

Postdoctoral Researcher

Luxembourg

Jan 2026 – Present

- Research in continual reinforcement learning for long-term space missions
- Mentorship of doctoral candidates in robot learning, simulation, and control

NASA Jet Propulsion Laboratory

Visiting Researcher

California, United States

Oct 2025 – Dec 2025

- Design of scalable high-fidelity simulation environments for future NASA JPL robots
- Simulated demonstration of continual reinforcement learning using world models
- Practical demonstration of sim-to-real transfer for autonomous crane logistics

University of Luxembourg

Doctoral Researcher

Luxembourg

Oct 2021 – Dec 2025

- Independent research in robot learning for space applications | 9 first-author publications
- Support and maintenance of laboratory infrastructure | LunaLab & Zero-G Lab facilities
- Teaching and mentorship of graduate students | 1 practical course & 2 MSc theses

Aivero

Rust Software Engineer (part-time)

Denmark

Jun 2019 – Aug 2021

- Development of real-time 3D video compression and streaming algorithms
- Integration and deployment of depth-sensing cameras on embedded hardware
- Migration of legacy C codebase to Rust for improved memory safety and concurrency

EDUCATION

University of Luxembourg

PhD in Space Robotics

Luxembourg

Oct 2021 – Dec 2025

Thesis: Robot Learning Beyond Earth: Enabling Adaptive Autonomy in Space

Aalborg University

MSc in Robotics

Denmark

Sep 2019 – Jun 2021

Thesis: Deep Reinforcement Learning for Robotic Grasping from Octrees

GPA: 11.5 / 12.0

Aalborg University

BSc in Robotics

Denmark

Sep 2016 – Jun 2019

Thesis: Eye Controlled Semi-Autonomous Assistive Manipulator

GPA: 11.4 / 12.0

TEACHING

University of Luxembourg

Robotic Manipulation in Space (Interdisciplinary Space Master)

Luxembourg

Sep 2022 – Present

- Design and delivery of practical sessions using MoveIt 2 on simulated and real robots

AWARDS

NASA Space ROS Sim Summer Sprint Challenge

Issued by NASA Tournament Lab

Online

Dec 2024

- Implementation of diverse Space ROS simulations | Individual submission — 2nd place (\$7,500 prize)

CERTIFICATIONS

IEEE CertifAIED™ Authorized Lead Assessor

Issued by IEEE Standards Association

Jun 2025

OPEN SOURCE PROJECTS

Space Robotics Bench

- Extensive simulation framework for robot learning research in the challenging domain of space

PyO3 Bindgen

- Compile-time generator of Rust FFI bindings to Python modules

PyMoveIt 2

- Accessible Python interface for MoveIt 2 via ROS 2 middleware

[GitHub Repositories]

SELECTED PUBLICATIONS

Advancing Adaptive Autonomy Through Procedural Space Environments

A. Orsula, M. Geist, M. Olivares-Mendez, C. Martinez

IAC 2025

Sep 2025

Learning Tool-Aware Adaptive Compliant Control for Autonomous Regolith Excavation

A. Orsula, M. Geist, M. Olivares-Mendez, C. Martinez

ASTRA 2025

Sep 2025

Sim2Dust: Mastering Dynamic Waypoint Tracking on Granular Media

A. Orsula, M. Geist, M. Olivares-Mendez, C. Martinez

iSpaRo 2025

Aug 2025

A Retrospective on the Robot Air Hockey Challenge

P. Liu, ..., A. Orsula, ..., J. Peters

NeurIPS 2024

Dec 2024

GraspLDM: Generative 6-DoF Grasp Synthesis using Latent Diffusion Models

K. R. Barad, A. Orsula, A. Richard, J. Dentler, M. Olivares-Mendez, C. Martinez

IEEE Access

Nov 2024

Leveraging Procedural Generation for Learning Autonomous Peg-in-Hole Assembly in Space

A. Orsula, M. Geist, M. Olivares-Mendez, C. Martinez

iSpaRo 2024

Jun 2024

Learning to Grasp on the Moon from 3D Octree Observations with Deep Reinforcement Learning

A. Orsula, S. Bøgh, M. Olivares-Mendez, C. Martinez

IROS 2022

Oct 2022

[Google Scholar Profile]

TECHNICAL SKILLS

Languages: English (professional), Danish (intermediate), Luxembourgish (basic), Slovak/Czech (native)

Robotics: Simulation, Perception, Planning, Control, Locomotion, Navigation, Manipulation, Compliance

Robot Learning: Reinforcement Learning (model-free and model-based), Imitation Learning, Sim-to-Real

Programming: Rust, Python, C, C++, CUDA, Shell, WASM, Linux, Git, Docker/Apptainer, CI/CD, Slurm

Tools: Isaac Sim, MuJoCo, Gazebo, ROS/ROS 2, PyTorch, JAX, TensorFlow, Blender, SolidWorks, Onshape